

Microcontrollers Reflection

In this module we experimented with the arduino nano esp32, pairing this technology with coding in micropython to eventually create and test a sensor that would detect differences in moisture. We began the module by working on simple breadboard design. We made a simple circuit with a light and a resistor and tested it under different voltage values. During this, I learned about the importance of creating resistor circuits in parallel. When we increased voltage, the resistor blew before the light did. We learned that you can use a resistor as a fail safe in order to protect more expensive pieces of your circuits. After this, we threw the arduino into our circuit and we used a light detection device to create a sensor. Using chatGPT to help code in micropython, we eventually developed a program that turns on the LEDs when the surrounding environment lacks light.

The next day we continued to use the arduino by pairing it with a motor and a switch. We coded the arduino so that when the switch is pressed, the motor would rotate 180 degrees. We then continued to use the arduino, pairing it with a moisture detector, and we set up a program in which the arduino would take 10 voltage measurements and save it to a CSV so that we could analyze the data later. We ran 5 tests with the moisture detector, testing different conditions every time. We found that the voltage decreased as we applied more moisture.

In this module, I learned the importance of nuance and specificity. When writing code, it's very important to have a clear goal defined before you begin coding so that you have a goal to work toward. From there, it is important to have an open mind while searching for a means to accomplish that goal. It takes a lot of trial and error in order to figure out how to get something done, but it is very rewarding to see your program or circuit work. This module was challenging,

but I had a lot of fun during the process of “figuring-out” and I look forward to seeing what the rest of FYRE has in store for me.